MURALI SAI BUDDAKKAGARI VENKATA

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SUMMARY

Applied AI Engineer specializing in LLM agent systems, multi-agent orchestration, and production eval pipelines. Built and shipped agentic workflows using LangGraph, LangChain, and GPT-4o - including a Human-in-the-Loop system completing ~85% of tasks autonomously with full audit trail. Experienced designing evaluation frameworks (RAGAS, faithfulness/relevancy metrics), analyzing failure modes, and iterating on prompts and retrieval strategies to measurably improve agent solve Rate. Researched domain-adaptive ensemble learning for medical nucleus detection (Faster R-CNN, Mask R-CNN, YOLO) at Stevens Institute of technology.

TECHNICAL SKILLS

- **Programming Languages**: Python, SQL, C++, PySpark, TypeScript
- **AI & Machine Learning**: Prompt Engineering & Design, Agent Evaluation (RAGAS), Failure Mode Analysis, Retrieval-Augmented Generation, Multi-Agent Systems, LangChain, LangGraph, LlamaIndex, CrewAI, HuggingFace, Transformers, PyTorch, TensorFlow, Model Context Protocol (MCP), FastAPI, ChromaDB, Scikit-learn, Embedding Models, LLMOps, LangSmith, Fine-tuning
- **Infrastructure & DevOps**: AWS (ECS Fargate, S3), Google Cloud Platform, BigQuery, Azure, Docker, PostgreSQL, GitHub Actions, CI/CD, ReactJS, OpenAI API, Claude API

EXPERIENCE

Querkey Inc — **Jun 2025 - Aug 2025**
Data Science Intern — *CA, US*

- Engineered end-to-end ML data ingestion pipelines (Python, PySpark, GCP, BigQuery) processing 200+ MP4 files daily, cutting validation latency by 40% and eliminating manual bottlenecks across 3 production data streams.
- Reduced production ML workflow failure rate by 60% by deploying a Dockerized observability stack (MLflow, HuggingFace monitoring, ChromaDB, FastAPI) over 100K+ records, enabling real-time failure detection and proactive incident response for business-critical pipelines.

VIEW Synergy — **Jan 2024 - Apr 2024**
AI and Automation Intern — *KA, IND*

- Reduced manual process overhead by ~35% for 100+ employees by architecting AI workflow automation pipelines (LangChain, LlamaIndex, Azure, ReactJS, TypeScript) enabling end-to-end intelligent task orchestration across full SDLC cycles.
- Shipped production RAG pipelines (C#/.NET 8.0, OpenAI API, Docker, SQL) with sub-200ms API latency, supporting reliable agentic tool-use in a regulated Agile delivery environment.

PROJECTS

Agent Orchestration System with Tool Use, Memory, and Human in the Loop

- Engineered a production-grade multi-agent system (LangGraph, GPT-4o, Claude Haiku) with 3-layer agent hierarchy, parallel task execution, and two-tier memory (Redis + ChromaDB), achieving ~85% autonomous task completion rate across long-horizon workflows.
- Designed a Human-in-the-Loop escalation system with 5 automated triggers and 4 approval levels, routing 15% of edge-case tasks to human reviewers with full audit trail - validated across fintech and legal compliance requirements.

SEC EDGAR RAG Filing Analyzer

- Improved RAG eval scores (faithfulness 0.57→0.65, relevancy 0.58→0.68, recall 0.57→0.70) across SEC 10-K filings by iteratively tuning retrieval strategies, chunk sizing, and context construction using LangChain, LlamaIndex, and ChromaDB.
- Deployed investment-banking-grade RAG system with RBAC, MNPI guardrails, and SEC 17a-4/FINRA 4511-aligned audit trail on AWS ECS Fargate (Terraform + GitHub Actions CI/CD) with an automated RAGAS eval pipeline, reducing manual eval effort by ~70% through continuous performance monitoring.

EDUCATION

Stevens Institute of Technology — **May 2026**
Master of Science, Applied Artificial Intelligence — *NJ, US*

PES University — **May 2024**
Bachelor of Technology, Computer Science (Specialization in ML) — *KA, IND*

PUBLICATIONS & CERTIFICATIONS

- Best Research Paper Award, ICICC 2024 - WorkFit: Real-time Multimodal Posture Monitoring System using CNN-based Computer Vision and Flex Sensor Fusion.